ELECTRONIC SPEED CONTROLLER

ARK 4IN1 ESC CONS

Connectorized NDAA-compliant 4-in-1 ESC running ARK32 firmware

The USA-built ARK 4IN1 ESC CONS is the connectorized variant of the ARK 4IN1 ESC. Four independent STM32F051 MCUs with TI DRV8328 gate drivers drive four 3-phase power stages under ARK32 firmware (supports AM32). Built-in motor and battery connectors eliminate soldering.

Board capability remains 50 A continuous / 75 A burst per motor; MR30 motor connectors limit practical continuous current to 30 A / 40 A burst. Includes 10 AWG power cables with M2.5 ring terminals.

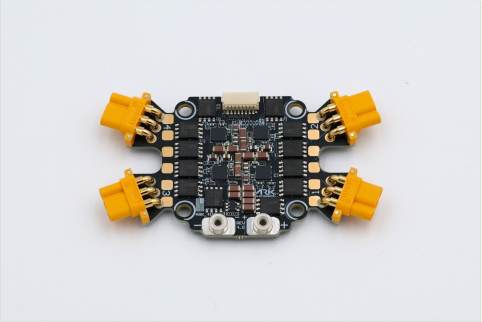
KEY FEATURES

- Board: 50 A continuous / 75 A burst per motor
- Connector limit: 30 A continuous / 40 A burst per motor
- 3–8S LiPo (6 V min, 65 V abs max)
- 4× independent STM32F051 channels
- Onboard current sensor (100 A/V)
- ARK32 firmware (supports AM32)
- No soldering: MR30 motor connectors, battery leads included

SPECIFICATIONS

ELECTRICAL	
Input voltage	3–8S LiPo, 6 V minimum
Absolute maximum	65 V
Board continuous	50 A per motor
Board burst	75 A per motor
Board total	About 170 A continuous
Connector continuous	30 A per motor (MR30)
Connector burst	40 A per motor
Current sense	100 A/V
Firmware	ARK32 (supports AM32)
Bulk capacitance	450 µF
Operating temperature	−20 °C to +85 °C
Storage temperature	−20 °C to +85 °C

MECHANICAL	
Dimensions	77.0 × 42.0 × 9.4 mm
Weight	24.0 g
Mounting	30.5 mm pattern



AT A GLANCE	
Voltage	3–8S LiPo
Board current	50 A / 75 A
Connector limit	30 A / 40 A
Firmware	ARK32 (AM32)
Sense	100 A/V
Dimensions	77 × 42 × 9.4 mm
Weight	24.0 g
Hardware rev	Rev 4.2

COMPLIANCE & SOURCING

- Designed and manufactured in the USA
- NDAA compliant
- Blue UAS Framework listed

ORDERING & RESOURCES

STORE arkelectron.com/product/ark-4in1-esc-cons/

DOCS docs.arkelectron.com/products/electronic-speed-controller/ark-4in1-esc

CAD github.com/ARK-Electronics/ark-hardware

FLIGHT CONTROLLER CONNECTOR

PIN	SIGNAL	NOTES
1	VBAT	Battery voltage
2	CURR	Current sense output
3	TELEM	Serial telemetry
4	MOTOR 1	DShot / PWM
5	MOTOR 2	DShot / PWM
6	MOTOR 3	DShot / PWM
7	MOTOR 4	DShot / PWM
8	GND	

Connector: 8-pin JST-SH.

DEBUG CONNECTOR

PIN	SIGNAL	NOTES
1	3.3 V	
2–9	SWDIO / SWDCLK 1–4	Per-motor SWD
10	GND	

Connector: 10-pin JST-SH.

NOTES

- Before first use, configure KV and pole count for each motor.
- DShot is the recommended input protocol. PWM requires calibration.
- Motor connections: 4× MR30. Battery: M2.5 threaded standoffs + included 150 mm 10 AWG cables with ring terminals.
- Board current ratings are thermal limits with airflow and have no fixed duration. All four channels share one current-sense shunt, which limits the board to about 170 A.
- The 450 µF of onboard bulk capacitance covers 3S–6S with short battery leads. At 8S, add an external bulk capacitor across the battery input to keep transients below the 65 V absolute maximum.

VARIANT

PRODUCT	KEY DIFFERENCE
ARK 4IN1 ESC	Solder-pad motors/battery; smaller 43 × 40.5 × 7.6 mm, 14.5 g; full 50 A / 75 A board rating

ARK-DS-ARK-4IN1-ESC-CONS | Version 1.0, 2026-10-29 | Compiled from the ARK 4IN1 ESC Production schematic Rev 4.2 (28 May 2026), the Production bill of materials, manufacturing documentation, and ARK Electronics published documentation. Board current ratings are thermal limits; connector ratings are additional limits of the MR30 connectors. Temperature ratings per component data sheets; the MR30 connectors set −20 °C. Specifications are subject to change without notice. © 2026 ARK Electronics LLC.